

$$\omega_{p,open} = \frac{1}{(r_{o2} \parallel r_{o4}) C_L}$$

$$A_{closed} = \frac{A_{open}}{1 + \beta A_{open}}$$

For this circuit $\beta = 1$

$$A_{closed} = \frac{A_{open}}{1 + A_{open}}$$

$$A_{open} = g_{mN} (r_{oN} \parallel r_{oP})$$

$$\approx g_{mN} \frac{r_{oN}}{2}$$

$$A_{open} \gg 1$$

In such case, $A_{closed} \approx 1$

$$R_{out,open} = r_{o2} \parallel r_{o4}$$

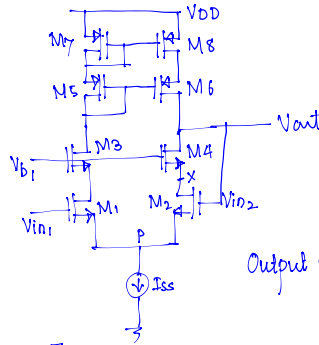
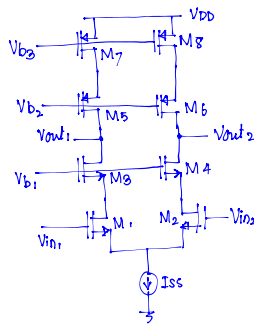
$$R_{out,closed} = \frac{R_{out,open}}{1 + \beta A_{open}} = \frac{R_{out,open}}{A_{open}}$$

$$= \frac{r_{o2} \parallel r_{o4}}{g_{m2} (r_{o2} \parallel r_{o4})}$$

$$= \frac{1}{g_{m2}}$$

$$\omega_{out,closed} = \frac{1}{\frac{1}{g_{m2}} C_L} = \frac{g_{m2}}{C_L}$$

Cascode Amplifiers:-



$$\text{Output voltage swing} = V_{DD} - \{ |V_{ov8}| + |V_{ov6}| + |V_{thp}| + V_{ov4} + V_{ov2} + V_{Iss} \}$$

$$A_{open} = g_{m1,2} [g_{m4} r_{o4} r_{o2} \parallel g_{m6} r_{o6} r_{o8}]$$

$$= g_{mN} [g_{mN} r_{oN}^2 \parallel g_{mp} r_{op}^2]$$

$$= g_{mN} \frac{g_{mN} r_{oN}^2}{2} = \left(\frac{g_{mN} r_{oN}}{2} \right)^2$$

$$g_{mp} = g_{mN}$$

$$r_{op} = r_{oN}$$

$$\text{Output voltage swing} = 2 \left[V_{DD} - \{ |V_{ov8}| + |V_{ov6}| + V_{ov4} + V_{ov2} + V_{Iss} \} \right]$$

For M4 to be in saturation

$$V_{out} - V_X \geq V_{b1} - V_X - V_{th4}$$

$$V_{out} \geq V_{b1} - V_{th4} \quad \text{--- (1)}$$

For M2 to be in saturation

$$V_X - V_P \geq V_{out} - V_P - V_{th2}$$

$$V_X + V_{th2} \geq V_{out}$$

$$V_{b1} - V_{GS4} + V_{th2} \geq V_{out} \quad \text{--- (2)}$$

$$V_{b1} - V_{GS4} + V_{th2} \geq V_{out} \geq V_{b1} - V_{th4}$$

